# Akshat Sharma

Fourth-year BS-MS student

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## Summary

I am a fourth-year BS-MS student at the Indian Institute of Science Education and Research (IISER) Kolkata whose research lies at the intersection of chemistry, statistical mechanics, information theory, and applied mathematics. My work focuses on developing theoretical frameworks for understanding molecular organization, self-assembly, complexity, and the emergence of living matter using chemically realizable self-assembling systems as experimentally accessible model platforms. My long-term objective is to uncover universal principles governing organization across chemistry, physics, and biology.

## Research Interests

**Core Areas:** Statistical Mechanics, Complex Systems, Information Theory, Biophysics, Self-Organization

## Education

**BS-MS Dual Degree** — **Indian Institute of Science Education and Research (IISER) Kolkata**, Chemical Sciences — West Bengal, India — 2023 - 2028
- National Level Exams Qualified: Joint Entrance Examination (Advanced) • Joint Entrance Examination (Mains) • IISER Aptitude Test (IAT)

**Senior Secondary School Certificate** — **Baluni Public School**, CBSE Class XII — Uttarakhand, India — 2022 - 2023
- Score: 88.8%

**Secondary School Certificate** — **Baluni Public School**, CBSE Class X — Uttarakhand, India — 2020 to 2021
- Score: 88.8%

## Experience

**Independent Research**, Independent Researcher — India — Jan 2026 - present
- Investigate a possible connection between thermodynamic variance extrema and complexity maximization in a confined logarithmic-potential model motivated by Manning condensation.
- Analysed lysozyme molecular-dynamics trajectories using GROMACS, tracking backbone RMSD and Ramachandran dihedral angles.
- Developed the Quantum Assembly Index (QAI), an information-theoretic descriptor of molecular organization derived directly from quantum electron density using QTAIM.
- Developed the Core–Halo Theorem, establishing a universal organizational structure for maximally complex probability distributions within generalized statistical complexity.
- Performed density functional theory (DFT) calculations using ORCA to investigate the electronic structure and bonding of homoatomic and heteroatomic Zintl clusters.

**IISER Kolkata**, Undergraduate Researcher — West Bengal, India
Supervisor: Dr. Soumyajit Roy — Dec 2023 - present

- Synthesized the giant polyoxometalate cluster $\{Mo_{132}\}$ keplerates and $\{Mo_{154}\}$ giant blue ring using self-assembly in aqueous medium and confirmed the formation using UV-vis spectroscopy and Raman spectroscopy.
- Conducted a detailed study of the polymorphism and solid-state phase transitions of acetaminophen (paracetamol).
- Synthesized metastable Paracetamol Form II (orthorhombic) and characterized the phase transformation using X-Ray Diffraction.
- Synthesized soft-oxometalates for $CO_2$ reduction and water oxidation catalysis.
- Analyzed crystal formation as a function of the crystallization medium and temperature by systematically varying ibuprofen concentration, DMSO content, and thermal conditions.
- Trained in advanced crystallographic refinement protocols (Olex2).

**IIT Ropar**, Research Intern — Punjab, India
Supervisor: Dr. C. M. Nagaraja — May 2025 - Jul 2025

- Gained foundational knowledge of Metal–Organic Frameworks (MOFs) and Covalent Organic Frameworks (COFs) through an extensive review of the scientific literature.
- Synthesized sterically demanding N-heterocyclic carbene (NHC) ligands and their organometallic complexes, confirming structural integrity via 1H NMR spectroscopy.

## Publications

**Quantifying the physical complexity of chemical architectures through electron-density topology** — Apr 2026
Introduces the Quantum Assembly Index (Aρ), an information-theoretic measure of molecular organization derived directly from quantum electron density topology.
Akshat Sharma
10.26434/chemrxiv.15000864/v4 (ChemRxiv)

**A Unified Discrete and Continuous Theory of Core–Halo Complexity Maximizers** — July 2026
Developed a unified variational framework for generalized statistical complexity, proving the universal emergence of core–halo organization in complexity-maximizing systems.
Akshat Sharma
arxiv.org/abs/2607.17907 (arXiv)

## Awards

**Reliance Foundation Scholarship** — 2023
Selected among the top 5,000 students across India for academic excellence.

## Skills

**Research Methods:** Statistical mechanics, Information-Theoretic analysis, Density Functional Theory, QTAIM, Molecular Dynamics, Python, Polyoxometalates, Self-Assembly, Single-Crystal Crystallography, Scientific Computing

## Positions of Responsibility

**IISER Kolkata Tennis Club**, Treasurer — West Bengal, India
Aug 2024 - Apr 2025

## Languages

**English:** Professional working proficiency

**Hindi:** Native speaker

## Relevant Coursework

**Chemistry**

**Mathematics**

**Computing**